Walmart (WMT) | Project 1 Validation and AI-Use

FIN 43900 | Purdue University | October 12, 2026

This report explains how I checked my Walmart financial models, what the tests showed, and how I used AI throughout the project. My goal was to make sure the results could be repeated while also identifying the assumptions that still needed more research.

1. How I Checked My Financial Model

After building my five-year model and DCF, I used several tests to see whether the calculations worked as expected. I kept the test results and supporting code in my GitHub repository.

Test	What I Checked	Recorded Result
Enterprise-to-Equity Bridge	A separate example with known inputs and an expected answer.	$800 million in equity value and $8.00 per share; PASS
Changed-Input Test (V2)	Lowered FY2027–FY2031 revenue growth by one percentage point each year.	Value decreased from $35.674421 to $30.667082 per share; PASS
Cold-Run Test	Ran seven Python scripts in fresh processes and checked protected files.	Seven successful script exits and 23 files unchanged; PASS
V3 Model and Scenarios	Checked modeled statements, financing, and DCF calculations.	Downside $32.40, base $36.78, upside $44.00; internal checks PASS

2. Enterprise-to-Equity Bridge Test

I tested the bridge separately from my Walmart model. I started with $1,000 million in enterprise value, added $100 million in cash, and subtracted $250 million in debt and $50 million in finance leases. This gave me $800 million in equity value. Dividing by 100 million shares gave me $8.00 per share, which matched the expected answer.

This confirmed that the bridge formula worked in the separate test. It did not confirm that every cash, debt, lease, or share-count input used in my Walmart valuation was correct.

3. Changed-Input Test

For this test, I recorded a prediction before running the code. I expected the valuation to decrease if revenue growth decreased. I then reduced the five annual revenue-growth assumptions by one percentage point in my V2 model, without changing the other independent inputs. The result decreased from $35.674421 to $30.667082 per share.

This test showed that the model reacted in the expected direction and that the changed inputs were isolated. I completed this test using V2, so the $35.67 baseline is different from the final V3 base-case

value of $36.78. The original locked record and test log are preserved in the repository.

4. Cold-Run and Reproducibility Results

On October 12, I ran python3 -B walmart_project1_cold_run.py. The script ran seven Python subprocesses, including the V3 financial model, DCF, scenarios, reverse DCF, comparable valuation, and two validation tests. Each process exited successfully. The log also showed matching before-and-after file hashes for 23 protected files.

I can use the same command again to check the model calculations. These results are helpful because someone reviewing my GitHub repository can repeat the tests. However, a successful run does not mean that every source value or forecast assumption has been independently verified.

5. Source Checks and Remaining Limitations

I also reviewed the financial inputs and definitions used in the model. Some reported balances matched the FY2026 filing, but the source review identified items that still needed attention, including how some receivables, supplier-finance obligations, and other working-capital accounts were classified.

My proposed September 23 valuation also uses July 31 balance-sheet amounts and an August 26 basic share count. I still need a consistent diluted share count and a more complete enterprise-to-equity review. The 7.00% WACC and 2.50% terminal growth assumption remain provisional, and around 82% of the base-case enterprise value comes from terminal value.

The upside case also includes approximately $9.022 billion of working-capital release in FY2027 without a separate operating plan supporting that improvement. These are reasons I would continue reviewing the model before treating the valuation as final.

6. How I Used AI

I used ChatGPT and Codex to help organize financial research, develop and review Python code, create tests, and draft parts of my supporting documentation. AI also helped with explaining results and preparing written materials for the project.

I ran the recorded tests and reviewed their results, but the test scripts and documentation were also developed with AI assistance. For the changed-input test, I recorded the prediction before running the test; AI helped refine the wording of the prediction beforehand. I have kept the locked record unchanged and documented this distinction in my AI-use record.

The final interpretation and investment recommendation are my responsibility. I kept my Watch–Defer recommendation because my modeled values were below the reported September 23 market price, while several assumptions still needed more verification.

7. What the Tests Showed Me

One of my biggest takeaways was that a model can run successfully without proving that every assumption is right. The bridge test, changed-input test, and cold run helped me check my calculations, but I still need to understand where the financial inputs come from and whether the assumptions make sense. That is why I kept the investment recommendation cautious.

Supporting Documents

The original test scripts, logs, source review, reproducibility report, locked changed-input record, and detailed AI-use record are available in my GitHub repository: WMT-Ticker---Research.

Locked Changed-Input Record — Supporting Exhibit

The original pre-run prediction must remain unchanged. The following is a separate post-test summary, not a replacement for the originally committed prediction. The original record is linked in the evidence section.

Test Identification

• Test name: Five-Year Revenue Growth Downside Test

• Date and time of prediction: October 12, 2026, 3:36 PM EDT

• Model and version: Walmart Project 1 V2 — Five-Year Three-Statement FCFF Model

• Baseline run/output reference: `project1_fcff_v2.py`; base-case valuation of $35.674421 per diluted share using a provisional 7.00% WACC and 2.50% terminal growth rate.

Human-Authored, AI-Off Prediction — Completed Before Running

• Input to change: Revenue Growth

• Original input and units: FY2027–FY2031 revenue growth rates of 4.3%, 4.2%, 4.0%, 3.8%, and 3.5%, respectively (annual percentage growth).

• Proposed new input and units: Decrease revenue growth by 1 percentage point each year.

• Expected output direction: Walmart's estimated value per share decreases.

• Why I expect that direction: Changes in projected cash flow can affect the present value of Walmart's future cash flows.

• Expected effect on the investment decision: I expect the lower valuation to strengthen the argument for keeping Walmart on the watchlist rather than investing at its current stock price.

Precommit Evidence

• Saved prediction timestamp: October 12, 2026, 3:39 PM EDT (user-reported)

• Git commit or other immutable pre-run evidence: https://github.com/olianaozbaki-ctrl/WMT-Ticker---Research/commit/3314c717bd983030ec2011fa99e60c406cd74e56

• Confirmation that prediction was written before execution: The prediction-only record was uploaded before the changed-input test, according to the recorded sequence (prediction at 3:36 PM, saved/uploaded at 3:39 PM, execution at 3:50 PM). The original commit should be retained unchanged as pre-run evidence; confirm its displayed GitHub timestamp independently.

Actual Results — Completed After Running

• Test execution date and time: October 12, 2026, 3:50 PM EDT (user-reported)

• Actual input change: FY2027–FY2031 revenue growth decreased by exactly 1 percentage point each year, from 4.3%, 4.2%, 4.0%, 3.8%, and 3.5% to 3.3%, 3.2%, 3.0%, 2.8%, and 2.5%. Other independent assumptions were held constant in the reported test.

• Actual output values and directions:

- Original base valuation: $35.674421 per diluted share.

- Changed-input valuation: $30.667082 per diluted share.

- Change in valuation: –$5.007339 per diluted share (approximately –14.04%).

- FY2027 FCFF change: –$468.710 million.

- FY2028 FCFF change: –$845.958 million.

- FY2029 FCFF change: –$1,257.139 million.

- FY2030 FCFF change: –$1,699.209 million.

- FY2031 FCFF change: –$2,162.975 million.

• Did the actual outcome match the prediction?: Yes. Walmart's estimated value per share decreased, matching the prediction. Annual FCFF also decreased in all five forecast years.

• Effect on the committee recommendation: The lower estimated intrinsic value reinforces the Watch–Defer recommendation rather than initiating a position at the comparison market price, subject to the remaining model limitations.

• Failure diagnosis or explanation of no change: No accounting or financing failure was reported. Both cases passed the model's reported checks. The changed-input case maintained minimum cash of approximately $8,626.0 million, with no revolver borrowing required. Lower forecast revenue growth reduced FCFF and the resulting valuation.

Evidence and Reproducibility

• Files used:

- `walmart_project1_proforma_v2.py`

- `project1_fcff_v2.py`

- `walmart_project1_revenue_growth_changed_input.py`

- `Walmart_Project1_Revenue_Growth_Changed_Input_Output.md`

- `Walmart_Project1_Locked_Changed_Input_Record.md`

• Test command: **[VERIFY AND ENTER EXACT COMMAND USED FOR ORIGINAL TEST RUN]**. If the script was executed as a standalone Python file, the expected command would be `python3 walmart_project1_revenue_growth_changed_input.py`, but do not claim this was the historical command without checking the execution log.

• Output log or screenshot: `Walmart_Project1_Revenue_Growth_Changed_Input_Output.md` and VS Code execution screenshot dated October 12, 2026.

• Remaining limitations: WACC, terminal growth, and enterprise-to-equity bridge assumptions remain provisional. Operating-account classifications, share-count definitions, market-date consistency, and some source inputs require final verification. Results in this record are those reported in the Codex test output and have not been independently reproduced here.